

Vulnerability Assessment Report

18th January 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2025 to August 2025. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The database server is valuable to the business because it stores sensitive information that must not be publicly accessible. Securing the data on the server is critical because threat actors could exploit vulnerabilities and cause a data breach if proper controls are not in place. If the server were compromised or disabled, the business could experience financial losses and a loss of customer trust due to exposed or unavailable data.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>E.g. Competitor</i>	<i>Obtain sensitive information via exfiltration</i>	1	3	3
<i>Threat actor</i>	<i>Obtain sensitive information via exfiltration</i>	2	3	6
<i>Disaster</i>	<i>Loss of data due to disasters</i>	3	3	9
<i>Insider</i>	<i>Obtain sensitive information since he knows where to find the data</i>	3	3	9

	<i>he needs for personal benefits</i>			
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Approach

The risks considered focus on the data storage and management methods used by the business. The likelihood of a threat occurring and the potential impact of each event were evaluated based on their effect on daily operations. A qualitative risk assessment approach was used to determine which threats pose the greatest risk to the organization. This method helps prioritize risks that could significantly impact business continuity and data security.

Remediation Strategy

The remediation strategy includes implementing authentication, authorization, and auditing mechanisms to ensure that only authorized users can access the database server. This includes enforcing strong passwords, role-based access control, and multi-factor authentication to limit user privileges. Data in transit should be encrypted using TLS instead of SSL. Additionally, IP allow-listing should be implemented to restrict database access to corporate office networks and prevent unauthorized external connections.